

Does Cost-Efficiency Travel?

All Findings and Results

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August 26, 2026

Abstract

This document collects every empirical finding from this paper in one place: what was measured, what the numbers are, and what they mean. It is written for readers who want the substance without the methodological scaffolding.

1 The Central Question

Agent leaderboards report task success and dollar cost. Practitioners pick a “cost-efficient” system from one benchmark and assume it stays efficient elsewhere. We test that assumption.

Three things get conflated in every unconditioned cost number: the model’s price, the benchmark’s difficulty, and the scaffold’s execution behavior. Disentangling them is the paper’s entire project.

2 What We Studied

2.1 Primary data

242 agent evaluations from the Holistic Agent Leaderboard (HAL): 9 benchmarks, 11 scaffolds, 29 model labels. Dollar cost and accuracy are available; tokens, retries, cache behavior, and per-task trajectories are not.

The only scaffold repeated broadly enough is the HAL Generalist Agent. Six benchmarks have enough shared labels for analysis: CORE-Bench Hard, GAIA, SciCode, ScienceAgentBench, SWE-bench Verified Mini, and TAU-bench Airline. This gives 86 rows and 25 distinct model labels.

2.2 Matched trace data

977 exact task-model pairs from Open-SWE-Traces (NVIDIA): MiniMax-M2.5 under two scaffolds (OpenHands and SWE-agent) on identical tasks. Resolved status, trajectory turns, tool calls, and message characters are available. Tokens and dollars are not.

2.3 Telemetry data

Two public coding-agent telemetry corpora: TraceLab (8,058 real sessions, 665,453 LLM rounds, 743,819 tool calls) and MIMO Claude Code (1,017 sessions, 4,690 rounds, one model). Neither has task outcomes or billing.

3 Finding 1: Raw Cost Ranks Transfer

Across the six high-overlap benchmark pairs, raw dollar-cost Spearman correlations range from 0.74 to 0.93, with a mean of 0.84. All survive permutation testing and Holm correction.

Benchmark pair	Shared labels	Spearman ρ
GAIA – SWE-mini	17	0.89
CORE – GAIA	14	0.93
CORE – TAU	14	0.88
GAIA – TAU	13	0.93
CORE – SWE-mini	12	0.74
SWE-mini – TAU	12	0.86

A practitioner using a GAIA-like leaderboard to rank raw spend for a SWE-like workload would get a useful prior.

4 Finding 2: But the Signal Is Mostly Label Propensity

A pooled model— $\log C_{i,b} = \mu + \alpha_i + \beta_b + \epsilon_{i,b}$, where α_i is a displayed-label fixed effect and β_b is a benchmark effect—explains 82.0% of observed cost variation (adjusted $R^2 = 72.7\%$) on 86 rows with 30 parameters.

After removing label and benchmark cost propensity, **no residual-cost rank association is significant** under a 20,000-draw permutation test.

The portable raw-cost signal is largely a stable observed label-level component. The public data cannot identify whether that component comes from provider pricing, reasoning settings, routing, token behavior, or cache treatment.

5 Finding 3: Success Adjustment Changes the Ranking

Aggregate dollars per expected success (CPS) reorders labels: mean CPS correlation is 0.59, compared with 0.84 for raw cost. Only three CPS associations survive permutation testing and Holm correction:

- CORE–GAIA: $\rho = 0.87$, 95% interval [0.63, 0.96]
- CORE–TAU: 0.83, [0.31, 1.00]
- GAIA–TAU: 0.78, [0.42, 0.94]

The clearest gap is GAIA–SWE-mini: raw cost exceeds CPS correlation by 0.60, with a 95% interval [0.14, 1.13]. A label that is predictably cheap is not necessarily predictably cheap per expected success on a different workload.

6 Finding 4: Held-Out Prediction Works for Four Workloads

Leave-one-benchmark-out (LOBO) prediction: learn each label’s cost propensity from the other workloads, predict the held-out cost rank.

Strong prediction:

- CORE: $\rho = 0.78$, $N = 14$, $p = 0.0016$
- GAIA: 0.87, $N = 16$, $p < 0.0001$

- SWE-mini: 0.90, $N = 16$, $p < 0.0001$
- TAU: 0.83, $N = 14$, $p = 0.0004$

Leave-one-label-out ranges remain positive for every omission.

Failed prediction:

- SciCode: unstable, $[-0.50, 0.14]$
- ScienceAgentBench: directionally negative, $[-0.77, -0.43]$

SciCode has very low mean accuracy (2.1%) and one-third zero-accuracy labels. ScienceAgentBench has the narrowest log-cost dispersion.

7 Finding 5: No Universal Cost-Accuracy Winner

No label tested on at least five benchmarks appears on every accuracy-cost frontier under any of three definitions: discrete nondominance, the HAL-inspired hull, or a 5% tolerance. The stronger claim that no label appears on more than 4/6 frontiers fails under tolerance: o4-mini Low reaches 5/6. We retain only the non-universal-winner conclusion.

8 Finding 6: Scaffolds Change Execution Burden

Across 977 matched task-model pairs (MiniMax-M2.5), OpenHands uses substantially fewer resources than SWE-agent at the same task:

Metric	SWE-agent median	OpenHands median	Ratio (OH/SA)
Trajectory turns	149	123	0.85 [0.82, 0.87]
Assistant turns / tool calls	74	61	0.85 [0.82, 0.86]
Total characters	194,263	176,708	0.95 [0.92, 0.98]

OpenHands is lower on 69.7% of pairs for turns and tool calls. Resolved status has no detected difference ($p = 0.197$).

9 Finding 7: The Failure-Cost Premium Is a Composition Artifact

This is the paper’s most counterintuitive finding.

The naive view (what prior work reports): Comparing each scaffold’s failed versus resolved trajectories across tasks, failures look 19–36% heavier. OpenHands turns ratio: 1.21 [1.07, 1.36]. SWE-agent trajectory characters: 1.33 [1.11, 1.54]. This matches the “expensive failure” claim in prior resource-effectiveness work.

The task-matched view (what happens when you hold the task fixed): Among 107 pairs where one scaffold resolved and the other failed on the *identical* task:

- The failed side is heavier only 47–52 times out of 107 on every metric ($p \geq 0.25$, two-sided binomial)
- After normalizing each trajectory by its own scaffold’s median burden, the failed/resolved ratio is 0.98–1.06 with intervals spanning 1
- 95% confidence intervals: [0.37, 0.57] raw, [0.43, 0.63] normalized

Power analysis: With 107 pairs, a two-sided binomial test at $\alpha = 0.05$ has 80% power to detect a premium fraction of 0.65 or larger (failed heavier on 70+ pairs). The observed 0.47–0.53

falls well below this threshold, ruling out strong premiums (>15 percentage points) but not modest ones.

Interpretation: Hard tasks make agents both burn more and fail. The observed cross-task “failure premium” is explained by which tasks failed, not by failure adding burden. Unconditioned failure-cost ratios measure workload composition, not a property of failing runs.

Scope: One model (MiniMax-M2.5), four-shard nonrandom sample, character/turn proxies not tokens or dollars.

10 Finding 8: Agent Inference Is Context-Dominated

Across two independent telemetry sources, observable inference volume is dominated by repeated carried context, not fresh generation.

TraceLab (real sessions):

- Median round: 132,092 input tokens, 249 output tokens
- Mean cache-read share: 93.7%
- Median session context growth: 47,267 tokens; P90 205,248
- Median final-round context: $3.6\times$ initial (P90 $9.9\times$)

MIMO Claude Code (one model):

- Corpus totals: 46.3M uncached input, 753.0M cache-read, 12.0M output tokens
- Cache-read volume is $16.3\times$ uncached input

A benchmark’s single dollar total hides whether spending comes from model price, cache policy, context growth, or fresh generation.

11 Finding 9: Error Burden Is Heavy Tailed

In TraceLab:

Observed tool errors	Median rounds	Median tool calls	Median summed input
0	9	14	0.34M
1	16	23	0.74M
2–5	30	39	1.63M
6+	130	146	14.65M

The top 1% of sessions hold 48.3% of summed input tokens. The top 10% hold 88.2%.

12 Finding 10: Error Spirals and Retry Behavior

Cascade: After a successful tool call, the next call errs 4.9% of the time. After an erroneous call, 21.0% ($4.29\times$ conditional increase).

Retry: 88.3% of errored tool calls are followed by another call to the same tool in the same session. 77.9% of those immediate retries succeed. Per-call recovery is common.

Spiral tails: Consecutive-error runs have median length 1 but P90 length 3 and a maximum of 368. Median successful run length is 9.

Cross-source replication: The MIMO corpus replicates the cascade ratio almost exactly: 4.3% after an error versus 1.0% after a success ($4.33\times$), despite a different model and source.

13 Finding 11: Tool Failures Are Slower and Less Informative

Across 743,819 TraceLab tool calls:

- Successful calls: median latency 140 ms, median result 602 characters
- Failed calls: median latency 313 ms, median result 208 characters
- P95 latency: 85.6s (failed) vs 15.8s (success)

Failures consume more wall time while yielding less information.

14 Finding 12: Large Tool Outputs Grow Context

When the current round contains under 1k tool-result characters, median next-input growth is 443 tokens. For 1k–10k characters it is 1,412 tokens; for over 10k it is 5,420 tokens. Tool verbosity is a concrete mechanism linking tool use to subsequent context burden.

15 Finding 13: MIMO Has a Data-Quality Issue

In every one of the 1,017 released MIMO sessions, the per-round usage block repeats one constant tuple: identical uncached input, cache creation, cache read, and output across all rounds of a session. Within-session token dynamics cannot be validated in this source.

The source is still useful for session counts, corpus totals, cross-session concentration (top 10% hold 37.9% of summed input), and content-derived error sequences. It is not usable for round-level growth analysis.

16 Finding 14: Public Telemetry Is Fragmented

No public source jointly exposes task, model, scaffold, outcome, tokens, cache use, retries, latency, and billing.

Source	Has	Lacks
HAL	Accuracy, dollar cost, task, model, scaffold	Tokens, traces, retries
TraceLab	Tokens, cache, tools, sessions	Task outcomes, dollars
MIMO	Tokens, cache, tools, sessions	Task outcomes, dollars, round fidelity
Open-SWE	Task, model, scaffold, resolved status	Tokens, dollars
Exgentic	Multi-benchmark usage, OTel spans	Rewards, task matching

This fragmentation motivates a minimum public telemetry standard for agent leaderboards.

17 All Figures

1. **Failure burden, discordant pairs** – Two panels: naive cross-task ratios (left) vs discordant-sign fractions (right). The premium vanishes at fixed task.

2. **Rank transfer, compact** – Pairwise Spearman correlations across benchmark pairs for raw cost and CPS.
3. **Raw cost vs cost per success** – Scatter showing how CPS reorders labels compared with raw cost.
4. **LOBO cost portability** – Predicted vs observed ranks for four held-out workloads.
5. **Matched trace burden** – OpenHands vs SWE-agent medians on log scale for turns, tool calls, and characters.
6. **Weak Pareto membership** – Per-label frontier presence across benchmarks.
7. **TraceLab tail and error burden** – Top-1%/10% concentration bar chart and error-bin burden scatter.
8. **Coverage heatmap** – Label presence across benchmarks and scaffolds.

18 Tests

25 tests pass across rank correlations, bootstrap intervals, CPS inference, Holm correction, LOBO prediction, and the end-to-end pipeline hash check. One warning (constant-input Spearman for extreme-pair CPS tests) is expected.

19 Summary

The paper’s thesis is now supported by three independent lines of evidence:

1. **Cost ranks transfer, but it is mostly label propensity.** The portable signal is real ($\rho = 0.74\text{--}0.93$) but largely absorbed by a fixed-effects model; residual ranks are insignificant.
2. **Success adjustment changes the ranking.** CPS is less portable (mean $\rho = 0.59$), and only three pairs survive correction.
3. **The failure-cost premium is a composition artifact.** At fixed task and model, failed trajectories carry the same burden as successful ones. Hard tasks make agents burn more and fail; unconditioned failure-cost ratios measure workload composition.

Every unconditioned aggregate—raw cost, success-adjusted efficiency, failure premiums—conflates which workloads were attempted with how agents behaved on them.